

$$\Delta L_k = qv$$

**11. Answer: B**

A and D are incorrect, because the astronaut is inside the gravitational field of the Earth, thus the Earth must act a gravitational force on the astronaut, which is his weight.

Since both the astronaut and the space capsules are moving in a circular motion around the Earth at the same radius and tangential velocity, they will have the same centripetal acceleration. Thus B is correct, while C is wrong because they have different masses.

**12. Answer: B**